BITS Pilani
Pilani | Dubai | Goa | Hyderabad

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in>

Final Lung cancer work description 24th Feb 2026

29 messages

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Tue, Feb 24, 2026 at 9:08 PM
To: ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>, Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>

Dear all
I have updated some functions for our work, this is the final update, and no changes will be made further. The task which you have to do this week is attached. Also, a new function list is added.

Please save the codes which you have done with previously given functions, as those will be used for comparative analysis. Please add the respective tables in the overleaf file. Complete the task by this saturday.

Let me know if you have any doubts. Also, please reply to this email thread only.

3 attachments

A_grouping_based_ensemble_methods_with_NSGA_II.pdf
65K

Group 1 bits pilani paper.pdf
2820K

DocScanner 24 Feb 2026 20-58.pdf
632K

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Sat, Feb 28, 2026 at 10:17 AM
To: ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Hello,
Hope you guys are working on that project! Waiting for the results.
[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Sun, Mar 1, 2026 at 11:33 AM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Hello Ma'am,

We are still working on it. We'll update you with the progress by tomorrow EOD.

Regards
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Wed, Mar 4, 2026 at 1:12 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Still waiting for EOD. Happy Holi!
[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Wed, Mar 4, 2026 at 1:16 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Hello Ma'am,

We ran out of credits while running the pipeline. We are in the process of figuring out a way out.

Also, wishing you a very Happy Holi.

Regards

[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Fri, Mar 6, 2026 at 9:40 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Any update on this?
[Quoted text hidden]

ADVIK KASHI VISHWANATH . <f20220973@hyderabad.bits-pilani.ac.in> Sat, Mar 7, 2026 at 2:13 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Dear ma'am,

Sorry for the delayed response, we will update the work by tomorrow.
[Quoted text hidden]

ADVIK KASHI VISHWANATH . <f20220973@hyderabad.bits-pilani.ac.in> Sat, Mar 7, 2026 at 2:28 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Dear Ma’am,

Following up on my previous email regarding the update. While running the computations, Table 16 ran into an out-of-memory (OOM) issue during execution, which prevented it from completing successfully.

However, the results for Tables 2, 3, 4, and 15 have been generated successfully. We have attached the screenshots of these results for your reference.

K-Fold Cross-Validation Results

Fold	Accuracy	Precision	Recall	F1	Specificity	Features
1	0.996	0.996	0.996	0.996	0.998	519.0
2	0.9993	0.9993	0.9993	0.9993	0.9997	521.0
3	0.9987	0.9987	0.9987	0.9987	0.9993	471.0
4	0.9987	0.9987	0.9987	0.9987	0.9993	499.0
5	0.9973	0.9973	0.9973	0.9973	0.9987	503.0
6	0.9967	0.9967	0.9967	0.9967	0.9983	514.0
7	0.998	0.998	0.998	0.998	0.999	538.0
8	0.9973	0.9973	0.9973	0.9973	0.9987	502.0
9	0.9973	0.9973	0.9973	0.9973	0.9987	518.0
10	0.9973	0.9973	0.9973	0.9973	0.9987	511.0
Mean	0.9977	0.9977	0.9977	0.9977	0.9988	509.6

Model Performance Summary

method	accuracy	precision	recall	f1	specificity	n_features
KNN	0.997	0.997	0.997	0.997	0.9985	510
SVM	0.9757	0.9757	0.9757	0.9757	0.9878	510
RF	0.9867	0.9867	0.9867	0.9867	0.9933	510
Ensemble (MV)	0.9917	0.9917	0.9917	0.9917	0.9958	510

Individual Classifier Performance

Classifier	Accuracy	Precision	Recall	F1	Specificity	Features
KNN	0.997	0.997	0.997	0.997	0.9985	510
SVM	0.9757	0.9757	0.9757	0.9757	0.9878	510
RF	0.9867	0.9867	0.9867	0.9867	0.9933	510
LR	0.97	0.97	0.97	0.97	0.985	510
XGBoost	0.993	0.993	0.993	0.993	0.9965	510

Comparison with Existing Methods

method	accuracy	precision	recall	f1	specificity	notes
Masud et al. (2021)	0.9633	0.965	0.9633	0.9633	0.9817	CNN+ML
Mangal et al. (2020)	0.97	0.97	0.97	0.97	0.985	CovidNet
Hatuwal & Thapa (2020)	0.9706	0.971	0.9706	0.9706	0.9853	CNN
Nishio et al. (2021)	0.95	0.953	0.95	0.95	0.975	Homology
Mehmood et al. (2022)	0.9773	0.978	0.9773	0.9773	0.9887	SBCNet
Talukder et al. (2022)	0.981	0.981	0.981	0.981	0.9905	ML+DL
Ours (KNN)	0.997	0.997	0.997	0.997	0.9985	DenseNet121+SE+AdaptGA

Fusion Method Performance

Method	Accuracy	Precision	Recall	F1	Specificity
Majority Vote	0.9917	0.9917	0.9917	0.9917	0.9958
Weighted Fusion	0.9947	0.9947	0.9947	0.9947	0.9973
GP1 (Max)	0.9967	0.9967	0.9967	0.9967	0.9983
GP2 (Product)	0.9953	0.9953	0.9953	0.9953	0.9977
GP3 (Ratio)	0.0	0.0	0.0	0.0	0.5
GP4 (Weighted Sum)	0.9953	0.9953	0.9953	0.9953	0.9977

We are currently working on resolving the memory issue for Table 16 and will update you once it is completed.
Thank you
[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Sun, Mar 8, 2026 at 4:56 AM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear Ma'am,

We have completed the experiments for Tables 2–15, and the results are attached in the zip file.

While running Table 16 (4 backbones + SE + NSGA-II + 5 classifiers + GP ensemble), we encountered an OOM issue. The NSGA-II feature selection step exceeds the limits of our current hardware (NVIDIA L4 GPU, 32 GB RAM, 8-core CPU). Even after reducing the population size, generations, and parallel jobs, the process still crashes.

Running Table 16 successfully would likely require a system with at least 64 GB RAM.

Regards

[Quoted text hidden]
--
Regards
Anshul Saxena
(2022B4A71041H)

result_tables.zip
1081K

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Thu, Mar 19, 2026 at 7:29 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

I think one of you is enrolled under CS dept. Please prepare the small report, may be 5 pages. Make sure it is plagiarism free. You can add results there.
[Quoted text hidden]

KRITII GUPTA . <f20220757@hyderabad.bits-pilani.ac.in> Fri, Mar 20, 2026 at 2:54 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear Ma'am,
I had converted the project into a CS project, and enrolled under the CS department. I will prepare the report and send it to you shortly.

Best regards,
Kritii
[Quoted text hidden]

ADVIK KASHI VISHWANATH . <f20220973@hyderabad.bits-pilani.ac.in> Fri, Mar 20, 2026 at 3:54 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear ma'am,

I have also enrolled as a CS design project.

I Just wanted to confirm who you were referring to since you mentioned only one person in the previous mail.

[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Fri, Mar 20, 2026 at 6:41 PM
To: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Advik and Kritii, you both have to write the report within 4-5 pages.
[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Mon, Mar 30, 2026 at 4:17 AM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Dear Ma’am,

I hope you are doing well.

I wanted to kindly check if there has been any update regarding the paper.

Please let me know if you would like any further clarifications from my side.

[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Tue, Mar 31, 2026 at 9:10 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Dear Students
Thanks for your mail. I was checking the results. Some queries are there:
1) Have you applied the new grouping function?
2) Most of the values in many tables are exactly the same, like there is not even a small change. Please check once!
3) It seems from the table that NSGA AND SIMPLE GA have not affected the results...Please check this again. Not even a small change!
4) Have the new grouping operators improved the results? Or were previous operators better?

I will create one overleaf file for you. You can put the tables in the NEW overleaf file.
[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Tue, Mar 31, 2026 at 9:27 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Dear Ma'am,

We'll go through our results once again and update you regarding the same.

Regards
Anshul Saxena
(2022B4A71041H)
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Tue, Mar 31, 2026 at 9:31 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

After you finalize the results, let's meet for discussion. Let me know when you are done from your side.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Sat, Apr 4, 2026 at 12:07 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>

Any update? We need to write this paper at the earliest.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Sat, Apr 4, 2026 at 1:48 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Lets meet on Tuesday via Google meet, from 2:30 pm for discussing the results. Let me know your availability.
[Quoted text hidden]

ADVIK KASHI VISHWANATH . <f20220973@hyderabad.bits-pilani.ac.in> Sun, Apr 5, 2026 at 11:18 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear ma'am,

Can we have the meet on Thursday, as we have a couple of evaluatives over the next 2-3 days.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Mon, Apr 6, 2026 at 10:35 PM
To: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Ok, let's meet on Thursday 4 pm.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Wed, Apr 8, 2026 at 8:12 PM
To: "ADVIK KASHI VISHWANATH ." <f20220973@hyderabad.bits-pilani.ac.in>
Cc: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>, KRITII GUPTA <f20220757@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Will you be available for online meet tomorrow at 4pm? You present your results in the meeting.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Wed, Apr 8, 2026 at 10:39 PM
To: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>, ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Ok, let's meet online tomorrow at 11pm.

On Wed, 8 Apr 2026, 20:38 KRITII GUPTA ., <f20220757@hyderabad.bits-pilani.ac.in> wrote:

> Dear Ma'am,
> Actually we have our farewell Batchsnaps tomorrow starting from 3.30 pm. Can we meet you tomorrow anytime in between 11 to 12 pm if possible?
>
> Best Regards,
> Kritii Gupta
> [Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Wed, Apr 8, 2026 at 10:46 PM
To: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>, ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Better if you all can prepare a ppt showing the modified algorithm and the results you got.

Also, let me know if anyone of you is available for the next semester project.
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Thu, Apr 9, 2026 at 8:10 AM
To: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>, ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Due to some urgent meetings, it's not possible to discuss with you all today. Let's meet tomorrow in the first half of the day. Please prepare a PPT for the same.
[Quoted text hidden]

KRITII GUPTA . <f20220757@hyderabad.bits-pilani.ac.in> Thu, Apr 9, 2026 at 2:37 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>

Cc: ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear Ma'am,
Sure we can meet tomorrow, do let us know what time is convenient for you. We have an extra class tomorrow from 10 to 11, so we are available anytime after 11 to show our results.

Ma'am on an informal note, could you tell us if you are coming for batch snaps today or tomorrow? We would love to take a picture with you.

Best regards,
Kritii Gupta
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Thu, Apr 9, 2026 at 8:28 PM
To: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>
Cc: ANSHUL SAXENA <f20221041@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear Kritii,

So you all are graduating this semester—wishing you very best for your future! It was really nice working with all of you. I won’t be able to come for the snaps as I have some other commitments. Do take lots of pictures and enjoy the moments.

Also, let’s try to complete the task within few days and communicate the paper at the earliest. I want to have a publication with you all.

Warm regards,
Swati

[Quoted text hidden]

ANSHUL SAXENA . <f20221041@hyderabad.bits-pilani.ac.in> Thu, Apr 9, 2026 at 8:30 PM
To: Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in>
Cc: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Dear Ma'am,

Thank you for the kind wishes! We’ll definitely miss you at the photo session.

We are also very eager to publish as soon as possible. We are working to finish the remaining tasks and will share the draft with you shortly.

Regards
Anshul Saxena
(2022B4A71041H)
[Quoted text hidden]

Swati Hait <swati.hait@hyderabad.bits-pilani.ac.in> Thu, Apr 9, 2026 at 11:29 PM
To: "ANSHUL SAXENA ." <f20221041@hyderabad.bits-pilani.ac.in>
Cc: "KRITII GUPTA ." <f20220757@hyderabad.bits-pilani.ac.in>, ADVIK KASHI VISHWANATH <f20220973@hyderabad.bits-pilani.ac.in>, Chandrima Debnath <chandrima5debnath@gmail.com>

Ok. I will send you Google meet link tomorrow. We will meet at 11:15 am online.
[Quoted text hidden]